

Saumik Chakraborty

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Professional Summary

Final-year Computer Science Engineering student with hands-on experience in **Python development, machine learning, and applied AI systems**. Strong background in **ML algorithms, data analysis, NLP, computer vision, and backend API development**. Experienced in building, evaluating, and deploying **data-driven and AI-powered applications** using FastAPI, Docker, and cloud platforms.

Technical Skills

Programming: Python, Java, SQL, NoSQL

Data Science: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Model Evaluation, Metrics

Machine Learning: Regression, Classification, Algorithm Selection, Hyperparameter Tuning, Model Validation

Deep Learning: CNNs, Transformers, YOLO, PyTorch, TensorFlow

NLP & LLMs: Text Classification, Embeddings, Semantic Search, RAG Pipelines, Prompt Engineering

Backend Development and Databases: FastAPI, Flask, REST APIs, Async APIs, MySQL, MongoDB

Libraries & Tools: NumPy, Pandas, Scikit-learn, Matplotlib

Cloud & DevOps: Docker, Render, AWS/GCP (foundational), CI/CD fundamentals

Version Control & OS: Git, Linux

Experience

AI / ML Intern — Nirveon X Omnicare Pvt. Ltd.

Feb 2025 – Sep 2025

- Developed **agentic AI and prompt-engineered chatbots** for mental health analysis using NLP and LLM-based workflows.
- Built **machine learning classification models** using game statistics and behavioral data, applying feature engineering and model evaluation.

AI Research Intern — Finessefleet Foundation

Jan 2025 – Jul 2025

- Implemented **embedding-based retrieval and reasoning pipelines** for AI-driven question answering systems.
- Built an **Indian regional language translation system** supporting **21 languages** using NLP preprocessing and model integration.

Projects

- Multi-Personality AI Chatbot (Deployed Backend) Live API**
 - Designed FastAPI-based LLM backend with persona routing and request validation
 - Deployed production-ready APIs with OpenAPI documentation on Render
- Multi-Agent Mathematics Solving Tutor System GitHub**
 - Designed multi-agent architecture with reasoning, verification, and explanation agents
 - Implemented Python-based agent orchestration for structured problem solving
- Order-to-Delivery Backend System Live API**
 - Built asynchronous FastAPI backend for order and inventory workflows
 - Implemented MongoDB persistence and environment-based configuration
- Job Fraud Detection System GitHub**
 - Developed NLP-based classification pipeline for fraud detection
 - Exposed trained models via Flask REST APIs for inference
- Human-Animal Detection System Live API**
 - Implemented YOLOv8-based computer vision inference using PyTorch
 - Supported real-time image and video prediction pipelines

Education

Bachelor of Technology in Computer Science and Engineering - NIST University (2022 - 2026)

CGPA: 8.09

Certifications

AWS Academy — Cloud Web Application Builder, ML for NLP, Generative AI

Google Data Analytics Professional Certificate

Google Cloud — Data Engineering and DevOps Foundations

IBM Watsonx Hackathon

Amazon ML Summer School 2024

Achievements

- 2nd Runner-Up** — Infosys Global Hackathon 2025
- Finalist** — Hack4Bihar National Innovation Challenge 2025